

Mitchelle Dery

7.24

```
MariaDB [(none)]> create database projects;  
Query OK, 1 row affected (0.017 sec)
```

```
MariaDB [(none)]> use projects  
Database changed
```

```
MariaDB [projects]> CREATE TABLE Employee (  
->     empNo      VARCHAR(10)  NOT NULL,  
->     fName     VARCHAR(20)  NOT NULL,  
->     lName     VARCHAR(20)  NOT NULL,  
->     sex       CHAR(1)      NOT NULL,  
->     position  VARCHAR(20)  NOT NULL,  
->     dob       DATE,  
->     salary    DECIMAL(10,2),  
->     PRIMARY KEY (empNo),  
->     CONSTRAINT chk_sex  
->         CHECK (sex IN ('M', 'F')),  
->     CONSTRAINT chk_position  
->         CHECK (position IN ('Manager', 'Team Leader',  
->                             'Analyst', 'Software Developer'))  
-> );
```

```
Query OK, 0 rows affected (0.069 sec)
```

```
MariaDB [projects]> CREATE TABLE Project (  
->     projNo     VARCHAR(10)  NOT NULL,  
->     projName   VARCHAR(50)  NOT NULL,  
->     startDate  DATE,  
->     endDate    DATE,  
->     budget     DECIMAL(10,2),  
->     mgrEmpNo   VARCHAR(10),  
->     PRIMARY KEY (projNo),  
->     FOREIGN KEY (mgrEmpNo) REFERENCES Employee(empNo)  
-> );
```

```
Query OK, 0 rows affected (0.038 sec)
```

```
MariaDB [projects]>
```

```
MariaDB [projects]> CREATE TABLE WorksOn (  
->     empNo      VARCHAR(10)  NOT NULL,  
->     projNo     VARCHAR(10)  NOT NULL,  
->     hoursWorked INT          NOT NULL,  
->     PRIMARY KEY (empNo, projNo),  
->     FOREIGN KEY (empNo) REFERENCES Employee(empNo),  
->     FOREIGN KEY (projNo) REFERENCES Project(projNo),  
->     CONSTRAINT chk_hours  
->         CHECK (hoursWorked BETWEEN 0 AND 40)  
-> );
```

```
Query OK, 0 rows affected (0.044 sec)
```

7.25

```
MariaDB [projects]> CREATE VIEW FemaleManagerProjects AS
```

```
-> SELECT p.projNo,  
->        p.projName,  
->        p.startDate,  
->        p.endDate,  
->        p.budget,  
->        p.mgrEmpNo  
-> FROM   Project p  
-> JOIN   Employee e ON p.mgrEmpNo = e.empNo  
-> WHERE  e.sex = 'F'  
-> ORDER BY p.projNo;
```

```
Query OK, 0 rows affected (0.048 sec)
```

7.26

```
MariaDB [projects]> CREATE VIEW EmpProjectHours AS
```

```
-> SELECT e.empNo,  
->        e.fName,  
->        e.lName,  
->        p.projName,  
->        w.hoursWorked  
-> FROM   Employee e  
-> JOIN   WorksOn w ON e.empNo = w.empNo  
-> JOIN   Project p ON w.projNo = p.projNo;
```

```
Query OK, 0 rows affected (0.025 sec)
```

7.27

```
MariaDB [projects]> CREATE VIEW EmpProject(empNo, projNo, totalHours)
```

```
-> AS SELECT w.empNo, w.projNo, SUM(hoursWorked)  
->    FROM   Employee e, Project p, WorksOn w  
->    WHERE  e.empNo = w.empNo AND p.projNo = w.projNo  
->    GROUP BY w.empNo, w.projNo;
```

```
Query OK, 0 rows affected (0.039 sec)
```

```
MariaDB [projects]> INSERT INTO Employee (empNo, fName, lName, sex,  
position, dob, salary)
```

```
-> VALUES  
->    ('E1', 'John', 'Smith', 'M', 'Manager',  
'1980-05-14', 55000.00),  
->    ('E2', 'Sarah', 'Johnson', 'F', 'Manager',  
'1985-03-22', 52000.00),  
->    ('E3', 'David', 'Brown', 'M', 'Team Leader',  
'1990-07-11', 45000.00),  
->    ('E4', 'Emily', 'Davis', 'F', 'Team Leader',  
'1992-01-30', 43000.00),
```

```

->      ('E5', 'James', 'Wilson', 'M', 'Analyst',
'1995-09-18', 38000.00),
->      ('E6', 'Linda', 'Taylor', 'F', 'Analyst',
'1993-11-25', 37000.00),
->      ('E7', 'Robert', 'Moore', 'M', 'Software
Developer','1997-04-09', 35000.00),
->      ('E8', 'Grace', 'Anderson','F', 'Software
Developer','1998-06-17', 34000.00);
Query OK, 8 rows affected (0.048 sec)
Records: 8  Duplicates: 0  Warnings: 0

```

```

MariaDB [projects]> INSERT INTO Project (projNo, projName, startDate,
endDate, budget, mgrEmpNo)
-> VALUES
->      ('P1', 'Database Upgrade', '2024-01-10', '2024-06-30',
80000.00, 'E1'),
->      ('P2', 'Web Portal', '2024-02-15', '2024-09-30',
120000.00, 'E2'),
->      ('SCCS', 'Security System', '2024-03-01', '2024-12-31',
95000.00, 'E1'),
->      ('P4', 'Mobile App', '2024-04-01', '2025-01-31',
150000.00, 'E2'),
->      ('P5', 'Data Analytics Tool', '2024-05-01', '2024-11-30',
60000.00, 'E1');
Query OK, 5 rows affected (0.002 sec)
Records: 5  Duplicates: 0  Warnings: 0

```

```

MariaDB [projects]> INSERT INTO WorksOn (empNo, projNo, hoursWorked)
-> VALUES
->      ('E1', 'P1', 20),
->      ('E1', 'SCCS', 15),
->      ('E2', 'P2', 30),
->      ('E2', 'P4', 25),
->      ('E3', 'P1', 40),
->      ('E3', 'P2', 35),
->      ('E4', 'SCCS', 38),
->      ('E4', 'P4', 20),
->      ('E5', 'P1', 30),
->      ('E5', 'P5', 40),
->      ('E6', 'P2', 25),
->      ('E6', 'SCCS', 30),
->      ('E7', 'P4', 40),
->      ('E7', 'P5', 35),
->      ('E8', 'P1', 15),
->      ('E8', 'P2', 20);
Query OK, 16 rows affected (0.005 sec)
Records: 16  Duplicates: 0  Warnings: 0

```

```
MariaDB [projects]> SELECT *
-> FROM EmpProject;
```

empNo	projNo	totalHours
E1	P1	20
E1	SCCS	15
E2	P2	30
E2	P4	25
E3	P1	40
E3	P2	35
E4	P4	20
E4	SCCS	38
E5	P1	30
E5	P5	40
E6	P2	25
E6	SCCS	30
E7	P4	40
E7	P5	35
E8	P1	15
E8	P2	20

16 rows in set (0.024 sec)

```
MariaDB [projects]> SELECT projNo
-> FROM EmpProject
-> WHERE projNo = 'SCCS';
```

projNo
SCCS
SCCS
SCCS

3 rows in set (0.006 sec)

```
MariaDB [projects]> SELECT COUNT(projNo)
-> FROM EmpProject
-> WHERE empNo = 'E1';
```

COUNT(projNo)
2

1 row in set (0.003 sec)

```
MariaDB [projects]> SELECT empNo, totalHours
-> FROM EmpProject
-> GROUP BY empNo;
```

```
+-----+-----+
```

empNo	totalHours
E1	20
E2	30
E3	40
E4	20
E5	30
E6	25
E7	40
E8	15

8 rows in set (0.002 sec)

7.28

The view stores a list of part numbers where the cost is greater than 1000 pounds. For insert, if a new part costs more than 1000 pounds, add it to the view. If it costs less do nothing. There is no need to reference the base table. For delete, you cannot just remove it from the view immediately. You must check if that part exists in another contract with a cost above 1000 pounds. If yes, keep it in the view but if no remove it. For update, if the cost goes above 1000 pounds, add it to the view and there is no need to reference base table Part. However, if the cost drops below 1000 pounds, check for other contracts before removing it from the view